

SatQuery AI — Analysis Report

Job job_1a90dc430a12 · task **sar_optical_fusion** · schema 2.0.0 · generated 2026-08-30T16:58:39.700176+00:00 · SIH problem statement 26167 (ISRO/SAC)

Query

Use the optical and SAR images together to identify built-up and water-covered regions.

Answer

Based on the measured signal, using the optical and SAR images jointly: built-up / man-made surface 10.7% (10.5 ha); water-covered 12.3% (12.1 ha); vegetation 24.1% (23.6 ha). Cross-modal agreement — water kappa 0.04 (optical 12.4%, SAR 1.3%); built-up / man-made sur.... Fusing the optical and sar channels, the built-up and water-covered extents are separated in the joint map below. [Mock output — SATQUERY MOCK is on; phrasing is template, not a measured model.]

Confidence

not available · source mock · calibrated: no

Mock backend was active (SATQUERY MOCK=1): no measurement was performed, so no confidence is reported.

Routing decision (the graded artefact)

field	value
classified task	sar_optical_fusion
classifier method	rules
plan	fusion.joint
registry entries used	fusion.joint
model / tool	mock-specialist
adapter	—
permitted parameters	min_region_pixels=64, smoothing_window=3, max_new_tokens=512, temperature=0.2, regions_per_class=4, speckle_filter=yes
rejected parameters	—
runner-up task	—
infeasible tasks	single_vqa, single_caption, single_grounding, change_vqa, change_description
narration source	mock
backend	mock
execution time	1374 ms

- Input is a co-registered optical+SAR pair, which is the defined input for cross-modal joint extraction.
- 'single_vqa' scored on wording but is infeasible: rs_vlm.vqa requires exactly one image, 2 supplied.
- Matched sar_optical_fusion evidence: optical and sar, together, sar.

Input validation and compatibility checks

check	status	message
format	passed	areaB_optical.tif: GEOTIFF, 512x512, 3 band(s), uint8
modality	passed	areaB_optical.tif: modality 'optical' (medium confidence) — 3 bands: true-colour optical composite
georeference	passed	areaB_optical.tif: CRS EPSG:32644, pixel size [1.9921875, 1.953125], extent [523000.0, 3120000.0, 524020.0, 3121000.0]
metadata	passed	areaB_optical.tif: acquisition date 2023-07-01 (from rasterio_tags)
format	passed	areaB_sar.tif: GEOTIFF, 512x512, 1 band(s), uint8
modality	passed	areaB_sar.tif: modality 'sar' (high confidence) — metadata/filename mentions 'sar'

check	status	message
georeference	passed	areaB_sar.tif: CRS EPSG:32644, pixel size [1.9921875, 1.953125], extent [523000.0, 3120000.0, 524020.0, 3121000.0]
metadata	passed	areaB_sar.tif: acquisition date 2023-07-01 (from rasterio_tags)
input_configuration	passed	Cross-modal pair detected: one optical/multispectral and one SAR image.
co_registration	passed	Footprints overlap 100.0% (native CRS EPSG:32644), resolution ratio 1.00x, identical CRS
temporal	passed	Cross-modal pair acquired 2023-07-01 / 2023-07-01.

Executed tool sequence

#	tool	parameters	outcome	ms	status
1	preprocessing.read_and_calibrate	file=areaB_optical.tif, speckle_filter=yes	optical · rgb · 512x512 analysis grid	49	ok
2	preprocessing.read_and_calibrate	file=areaB_sar.tif, speckle_filter=yes	sar · sar_single · 512x512 analysis grid	288	ok
3	preprocessing.modality_split	—	optical=areaB_optical.tif (rgb), sar=areaB_sar.tif (sar_single)	0	ok
4	analysis.optical_sar_fusion	min_region_pixels=64, smoothing_window=3, regions_per_class=4	mean cross-modal kappa=0.0184; joint water=0.123, built-up=0.107; obscured=0.000	777	ok
5	render.fusion_map	—	job_1a90dc430a12_fusion.png, job_1a90dc430a12_fusion_pair.png	142	ok
6	fusion.joint	min_region_pixels=64, smoothing_window=3, max_new_tokens=512, temperature=0.2, regions_per_class=4, speckle_filter=yes	464 chars, 7 box(es), mask=fusion_map, narration=mock	921	ok

Measurements — sar_optical_fusion (dual_branch_threshold_agreement_v1)

class	extent	area (ha)	regions	reliability	evidence basis
unclassified	52.8%	51.64	0	medium	residual after joint assignment
vegetation	24.1%	23.59	0	medium	optical only: the SAR product carries no discriminator for this class (single-polarisation)
water	12.3%	12.06	0	high	cross-modal: pixels confirmed by both sensors, plus SAR-only detections where the optical scene is obscured, plus optical-only detections in clear areas
built-up / man-made surface	10.7%	10.51	0	high	cross-modal: pixels confirmed by both sensors, plus SAR-only detections where the optical scene is obscured, plus optical-only detections in clear areas

class	optical	SAR	joint	IoU	Cohen kappa	SAR-only
water	12.4%	1.3%	12.3%	0.031	0.0366	0.9%
built_up	10.8%	0.0%	10.7%	0.0001	0.0002	0.0%
vegetation	24.1%	—	24.1%	—	—	0.0%

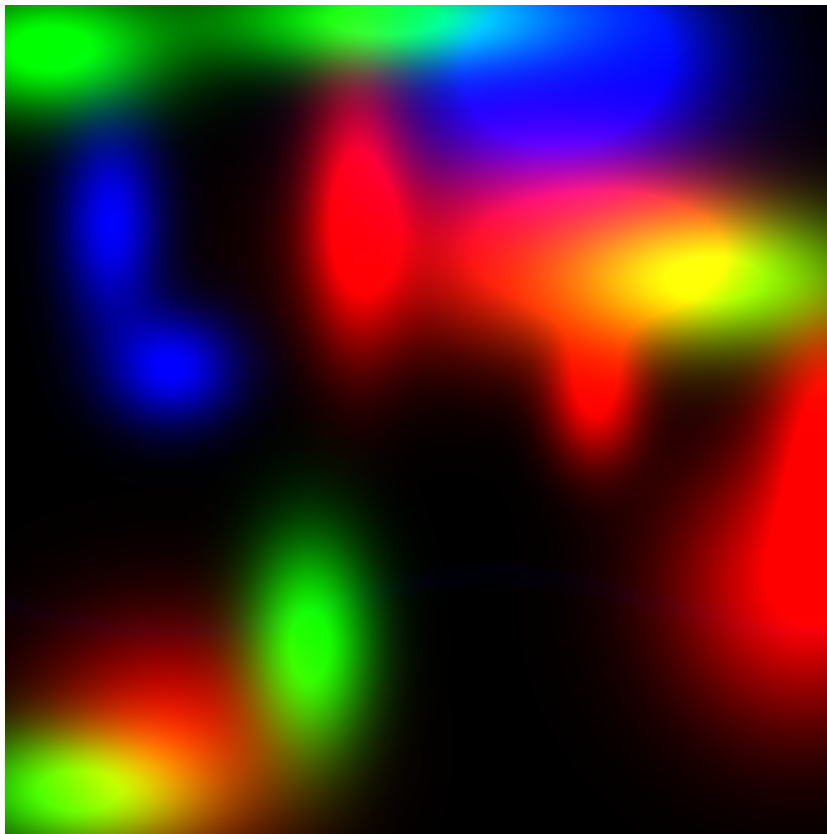
label	box [x1,y1,x2,y2] 0-100	position	area (ha)	extent	score
water	0.2, 6.05, 34.38, 58.59	west	8.822	9.0%	0.751
built-up / man-made surface	67.38, 0.2, 90.82, 27.34	north-east	2.052	2.1%	0.602
water	87.3, 0.2, 99.8, 18.16	north-east	1.951	2.0%	0.815
built-up / man-made surface	51.37, 29.3, 78.71, 52.73	centre	1.876	1.9%	0.588
built-up / man-made surface	6.45, 79.69, 32.42, 99.8	south-west	1.823	1.9%	0.609
water	43.36, 59.77, 67.97, 74.22	south	1.291	1.3%	0.608
built-up / man-made surface	36.91, 8.01, 49.61, 39.26	north	1.261	1.3%	0.59

Stated limitations

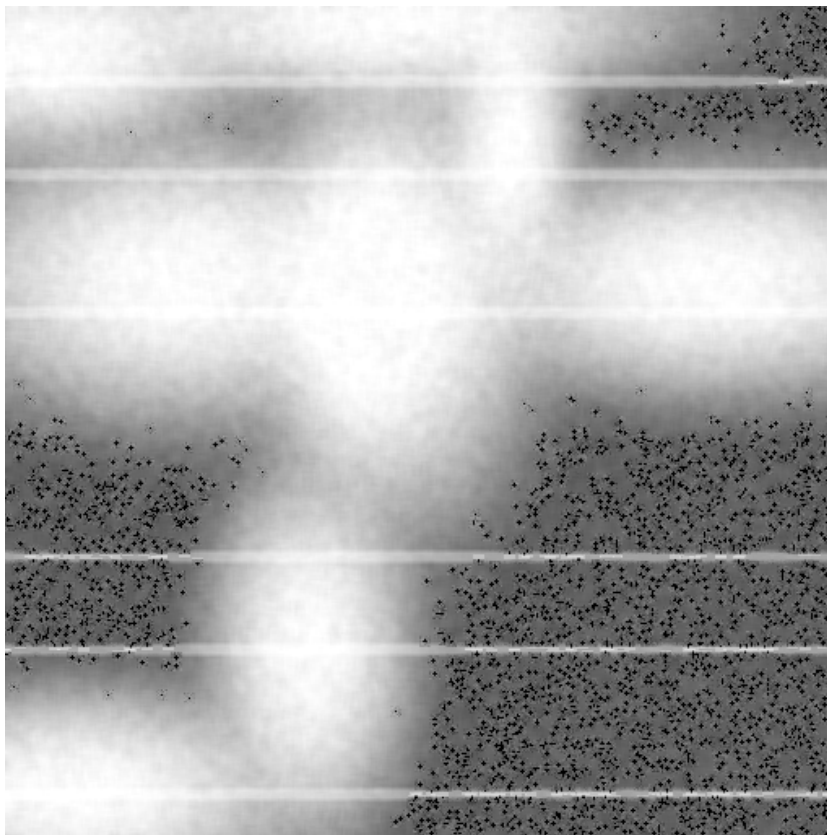
- No near-infrared band: NDVI/NDWI unavailable, vegetation and water detected from visible-band proxies (ExG, blue-over-red) with reduced reliability.
- SAR-only input: classes derive from backscatter level and roughness; vegetation vigour and water chemistry are not observable.
- 0.0% of the optical scene is bright, smooth and non-vegetated (brightness ≥ 0.85): treated as cloud/haze obscured

Unavailable indices: MNDWI, NDBI, NDVI, NDWI

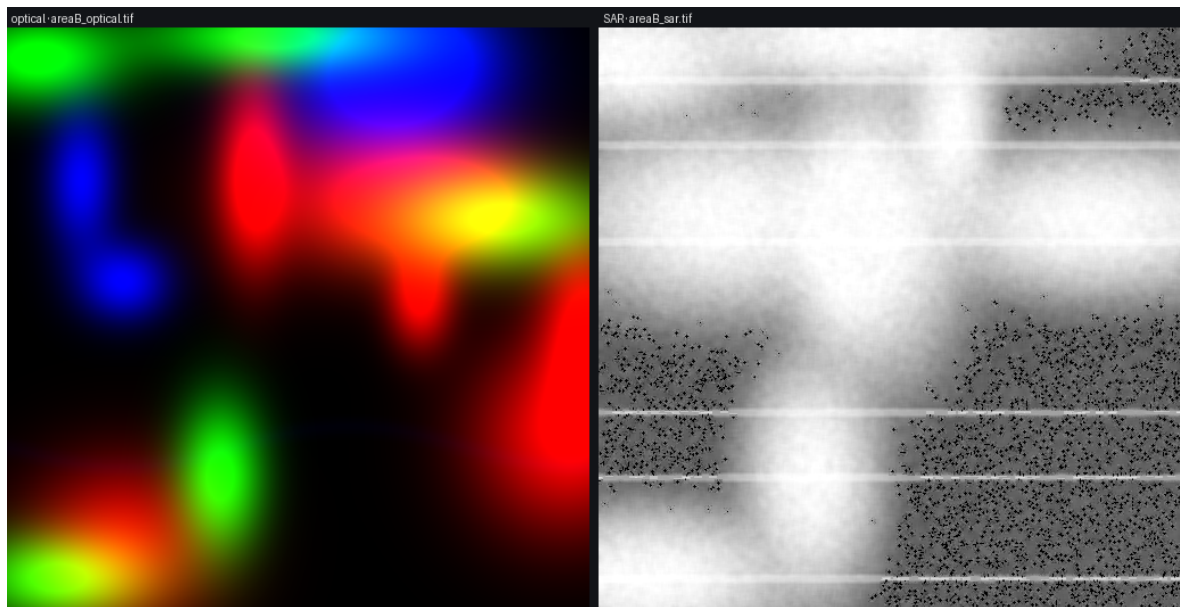
Visual evidence



input_preview — optical · areaB_optical.tif



input_preview — SAR · areaB_sar.tif (amplitude)



side_by_side — Co-registered optical / SAR pair after SAR calibration



fusion_map — Joint optical+SAR classes; white outline marks optically obscured area

Audit trace (evaluated artefact, verbatim)

```
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          81.24478429825919,
          28.214624141109027
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```

```

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      "Refined-Lee speckle filter applied (7x7, edge-preserving).",
      "Backscatter spans -17.6..45.9 dB, outside the calibrated sigma0 range: treated as an uncalibrated digital-number product, so level-based threshold
      itself rather than from published physical values."
    ]
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  "AMPLITUDE: Estimated ENL 2.6 from the most homogeneous windows.",
  "AMPLITUDE: Refined-Lee speckle filter applied (7x7, edge-preserving).",
  "AMPLITUDE: Backscatter spans -17.6..45.9 dB, outside the calibrated sigma0 range: treated as an uncalibrated digital-number product, so level-based threshold
  itself rather than from published physical values."
]
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    "areaB_optical.tif: acquisition date 2023-07-01 (from rasterio.tags)",
    "areaB_sar.tif: GEOTIFF, 512x512, 1 band(s), uint8",
    "areaB_sar.tif: modality 'sar' (high confidence) \u2014 2014 metadata/filename mentions 'sar'",
    "areaB_sar.tif: CRS EPSG:32644, pixel size [1.9921875, 1.953125], extent [523000.0, 3120000.0, 524020.0, 3121000.0]",
    "areaB_sar.tif: acquisition date 2023-07-01 (from rasterio.tags)",
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    "Footprints overlap 100.0% (native CRS EPSG:32644), resolution ratio 1.00x, identical CRS",
    "Cross-modal pair acquired 2023-07-01 / 2023-07-01."
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    "modalities": [
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      "sar"
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  }
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    "same_crs": true,
    "overlap_ratio": 1.0,
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  "ambiguous": false,
  "notes": [
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      "notes": [
        "Interpreted as amplitude; converted to dB.",
        "Estimated ENL 2.6 from the most homogeneous windows.",
        "Refined-Lee speckle filter applied (7x7, edge-preserving).",
        "Backscatter spans -17.6..45.9 dB, outside the calibrated sigma0 range: treated as an uncalibrated digital-number product, so level-based thresh
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      "AMPLITUDE: Estimated ENL 2.6 from the most homogeneous windows.",
      "AMPLITUDE: Refined-Lee speckle filter applied (7x7, edge-preserving).",
      "AMPLITUDE: Backscatter spans -17.6..45.9 dB, outside the calibrated sigma0 range: treated as an uncalibrated digital-number product, so level-bas
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    "sar"
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  "accepted": true
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"classification": {
  "task": "sar_optical_fusion",
  "method": "rules",
  "reasons": [
    "Input is a co-registered optical+SAR pair, which is the defined input for cross-modal joint extraction.",
    "'single_vqa' scored on wording but is infeasible: rs_vlm.vqa requires exactly one image, 2 supplied.",
    "Matched sar_optical_fusion evidence: optical and sar, together, sar."
  ],
  "scores": {
    "single_vqa": -1.0,
    "single_caption": -1.0,
    "single_grounding": -1.0,
    "change_vqa": -1.0,
    "change_description": -1.0,
    "sar_optical_fusion": 8.8
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  "alternatives": {
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    "single_caption": "rs_vlm.caption requires exactly one image, 2 supplied",
    "single_grounding": "rs_vlm.grounding requires exactly one image, 2 supplied",
    "change_vqa":
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  "change_description_requires_two_acquisitions_of_the_same_modality": "change.description requires two acquisitions of the same modality; an optical+SAR pair was supplied"
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"candidate_tasks": [
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  "change_description",
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]

```

```

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"rejected_parameters": []
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    "detail": {
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      "pixel_area_m2": 3.8909912109375
    }
  },
  {
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    "tool": "preprocessing.read_and_calibrate",
    "task": "",
    "parameters": {
      "file": "areaB_sar.tif",
      "speckle_filter": true
    },
    "outputs_summary": "sar \u00b7 sar_single \u00b7 512x512 analysis grid",
    "duration_ms": 288,
    "status": "ok",
    "detail": {
      "band_roles": {
        "amplitude": 0
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      "pixel_area_m2": 3.8909912109375
    }
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  {
    "order": 3,
    "tool": "preprocessing.modality_split",
    "task": "sar_optical_fusion",
    "parameters": {},
    "outputs_summary": "optical=areaB_optical.tif (rgb), sar=areaB_sar.tif (sar_single)",
    "duration_ms": 0,
    "status": "ok",
    "detail": {
      "optical_bands": [
        "blue",
        "green",
        "red"
      ],
      "sar_channels": [
        "amplitude"
      ],
      "sar_radiometry": {
        "amplitude": "amplitude"
      },
      "sar_enl": {
        "amplitude": 2.57
      },
      "speckle_filtered": true
    }
  },
  {
    "order": 4,
    "tool": "analysis.optical_sar_fusion",
    "task": "sar_optical_fusion",
    "parameters": {
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      "smoothing_window": 3,
      "regions_per_class": 4
    },
    "outputs_summary": "mean cross-modal kappa=0.0184; joint water=0.123, built-up=0.107; obscured=0.000",
    "duration_ms": 777,
    "status": "ok",
    "detail": {
      "agreement": [
        {
          "class_name": "water",
          "optical_fraction": 0.12355,
          "sar_fraction": 0.01349,
          "joint_fraction": 0.12334,
          "iou": 0.031,

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    "cohen_kappa": 0.0366,
    "sar_only_fraction": 0.00938,
    "optical_only_fraction": 0.11943
  },
  {
    "class_name": "built_up",
    "optical_fraction": 0.1077,
    "sar_fraction": 1e-05,
    "joint_fraction": 0.10749,
    "iou": 0.0001,
    "cohen_kappa": 0.0002,
    "sar_only_fraction": 0.0,
    "optical_only_fraction": 0.10769
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  {
    "class_name": "vegetation",
    "optical_fraction": 0.24124,
    "sar_fraction": null,
    "joint_fraction": 0.24121,
    "iou": null,
    "coh
en_kappa": null,
    "sar_only_fraction": 0.0,
    "optical_only_fraction": 0.24124
  }
],
"complementarity": [
  "Optical and SAR disagree about water (kappa 0.04): reported as low-agreement rather than merged.",
  "Optical and SAR disagree about built-up / man-made surface (kappa 0.00): reported as low-agreement rather than merged."
]
},
{
  "order": 5,
  "tool": "render.fusion_map",
  "task": "sar_optical_fusion",
  "parameters": {},
  "outputs_summary": "job_1a90dc430a12_fusion.png, job_1a90dc430a12_fusion_pair.png",
  "duration_ms": 142,
  "status": "ok",
  "detail": {}
},
{
  "order": 6,
  "tool": "fusion.joint",
  "task": "sar_optical_fusion",
  "parameters": {
    "min_region_pixels": 64,
    "smoothing_window": 3,
    "max_new_tokens": 512,
    "temperature": 0.2,
    "regions_per_class": 4,
    "speckle_filter": true
  },
  "outputs_summary": "464 chars, 7 box(es), mask=fusion_map, narration=mock",
  "duration_ms": 921,
  "status": "ok",
  "detail": {
    "reason_in_plan": "registry entry for classified task 'sar_optical_fusion'",
    "primary": true
  }
}
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"effective_parameters": {
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  "smoothing_window": 3,
  "max_new_tokens": 512,
  "temperature": 0.2,
  "regions_per_class": 4,
  "speckle_filter": true
},
"measurements": {
  "sar_optical_fusion": {
    "kind": "fusion",
    "method": "dual_branch_threshold_agreement_v1",
    "joint_classes": [
      {
        "name": "other",
        "label": "unclassified",
        "pixel_count": 132725,
        "fraction": 0.5279666174733182,
        "area_ha": 51.64318084716797,
        "evidence_basis": "residual after joint assignment",
        "reliability": "medium",
        "region_count": 0
      },
      {
        "name": "vegetation",
        "label": "vegetation",
        "pixel_count": 60637,
        "fraction": 0.2412078491899009,
        "area_ha": 23.59380340576172,
        "evidence_basis": "optical only: the SAR product carries no discriminator for this class (single-polarisation)",
        "reliability": "medium",
        "region_count": 0
      },
      {
        "name": "water",
        "label": "water",
        "pixel_count": 31006,
        "fraction": 0.12333873001603093,

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    "area_ha": 12.064407348632812,
    "evidence_basis": "cross-modal: pixels confirmed by both sensors, plus SAR-only detections where the optical scene is obscured, plus optical-only de
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    "region_count": 0
  },
  {
    "name": "built_up",
    "label": "built-up / man-made surface",
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    "fraction": 0.10748680332074992,
    "area_ha": 10.513847351074219,
    "evidence_basis": "cross-modal: pixels confirmed by both sensors, plus SAR-only detections where the optical scene is obscured, plus optical-only de
    "reliability": "high",
    "region_count": 0
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],
"agreement": [
  {
    "class_name": "water",
    "optical_fraction": 0.12355,
    "sar_fraction": 0.01349,
    "joint_fraction": 0.12334,
    "iou": 0.031,
    "cohen_kappa": 0.0366,
    "sar_only_fraction": 0.00938,
    "optical_only_fraction": 0.11943
  },
  {
    "class_name": "built_up",
    "optical_fraction": 0.1077,
    "sar_fraction": 1e-05,
    "joint_fraction": 0.10749,
    "iou": 0.0001,
    "cohen_kappa": 0.0002,
    "sar_only_fraction": 0.0,
    "optical_only_fraction": 0.10769
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  {
    "class_name": "vegetation",
    "optical_fraction": 0.24124,
    "sar_f
raction": null,
    "joint_fraction": 0.24121,
    "iou": null,
    "cohen_kappa": null,
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    "optical_only_fraction": 0.24124
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],
"mean_agreement": 0.0184,
"obscured_fraction": 0.0,
"complementarity": [
  "Optical and SAR disagree about water (kappa 0.04): reported as low-agreement rather than merged.",
  "Optical and SAR disagree about built-up / man-made surface (kappa 0.00): reported as low-agreement rather than merged."
],
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    "class_name": "water",
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      6.05,
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    ],
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      31,
      176,
      300
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    ],
    "area_px": 22673,
    "area_ha": 8.822,
    "fraction": 0.09019,
    "position": "west",
    "compactness": 0.482,
    "score": 0.751
  },
  {
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    "class_name": "built_up",
    "bbox_norm": [
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      0.2,
      90.82,
      27.34
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    "bbox_px": [
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      1,
      465,
      140
    ],
    "centroid_norm": [
      80.69,
      16.04
    ],
    "

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"area_px": 5273,
"area_ha": 2.052,
"fraction": 0.02098,
"position": "north-east",
"compactness": 0.316,
"score": 0.602
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{
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  "class_name": "water",
  "bbox_norm": [
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    0.2,
    99.8,
    18.16
  ],
  "bbox_px": [
    447,
    1,
    511,
    93
  ],
  "centroid_norm": [
    93.97,
    8.67
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  "area_px": 5014,
  "area_ha": 1.951,
  "fraction": 0.01995,
  "position": "north-east",
  "compactness": 0.852,
  "score": 0.815
},
{
  "label": "built-up / man-made surface",
  "class_name": "built_up",
  "bbox_norm": [
    51.37,
    29.3,
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    52.73
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    263,
    150,
    403,
    270
  ],
  "centroid_norm": [
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    40.58
  ],
  "area_px": 4821,
  "area_ha": 1.876,
  "fraction": 0.01918,
  "position": "centre",
  "compactness": 0.287,
  "score": 0.588
},
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  "class_name": "built_up",
  "bbox_norm": [
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    79.69,
    32.42,
    99.8
  ],
  "bbox_px": [
    33,
    408,
    166,
    511
  ],
  "centroid_norm": [
    20.2,
    87.9
  ],
  "area_px": 4685,
  "area_ha": 1.823,
  "fraction": 0.01864,
  "position": "south-west",
  "compactness": 0.342,
  "score": 0.609
},
{
  "label": "water",
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  "bbox_norm": [
    43.36,
    59.77,
    67.97,
    74.22
  ],
  "bbox_px": [
    222,
    306,
    348,
    380
  ],
  "centroid_norm": [
    53.55,

```

```

    67.97
  ],
  "area_px": 3319,
  "area_ha": 1.291,
  "fraction": 0.0132,
  "position": "south",
  "compactness": 0.356,
  "score": 0.608
},
{
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  "class_name": "built_up",
  "bbox_norm": [
    36.91,
    8.01,
    49.61,
    39.26
  ],
  "bbox_px": [
    189,
    41,
    254,
    201
  ],
  "centroid_norm": [
    41.17,
    26.36
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  "area_px": 3241,
  "area_ha": 1.261,
  "fraction": 0.01289,
  "position": "north",
  "compactness": 0.312,
  "score": 0.59
}
],
"optica
l": {
  "kind": "land_cover",
  "method": "visible_proxy_thresholds_v1",
  "classes": [
    {
      "name": "water",
      "label": "water",
      "pixel_count": 31252,
      "fraction": 0.12330102066984665,
      "area_ha": 12.160125732421875,
      "evidence_basis": "visible-only proxy: (Blue-Red)/(Blue+Red) > -0.066 (otsu) and brightness below p45",
      "reliability": "medium",
      "region_count": 3
    },
    {
      "name": "vegetation",
      "label": "vegetation",
      "pixel_count": 60663,
      "fraction": 0.2393385964704629,
      "area_ha": 23.603919982910156,
      "evidence_basis": "visible-only proxy: excess-green > +0.020 (otsu_rejected_prior_used)",
      "reliability": "medium",
      "region_count": 4
    },
    {
      "name": "built_up",
      "label": "built-up / man-made surface",
      "pixel_count": 27158,
      "fraction": 0.10714863430665862,
      "area_ha": 10.567153930664062,
      "evidence_basis": "brightness > p60 (0.218) with high 5x5 texture > p65 (0.006)",
      "reliability": "medium",
      "region_count": 12
    },
    {
      "name": "bare_soil",
      "label": "bare soil or fallow ground",
      "pixel_count": 95969,
      "fraction": 0.3786341882972134,
      "area_ha": 37.341453552246094,
      "evidence_basis": "smooth, non-green, non-water residue",
      "reliability": "medium",
      "region_count": 13
    },
    {
      "name": "other",
      "label": "unclassified",
      "pixel_count": 38419,
      "fraction": 0.15157756025581845,
      "area_ha": 14.948799133300781,
      "evidence_basis": "no proxy threshold satisfied",
      "reliability": "medium",
      "region_count": 12
    }
  ],
  "dominant": "bare_soil",
  "indices": [
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      "name": "BRIGHTNESS",
      "available": true,
      "mean": 0.1912,
      "std": 0.1671,
      "p05": 0.0039,
      "p95": 0.5041,

```

```

    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "mean scaled reflectance across available bands"
  },
  {
    "name": "BWI",
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    "mean": -0.4034,
    "std": 0.8124,
    "p05": -1.0,
    "p95": 1.0,
    "threshold": -0.0664,
    "threshold_method": "otsu",
    "separability": 0.9104,
    "note": "data-driven split inside the plausible BWI window [-0.10, +0.14]"
  },
  {
    "name": "EXG",
    "available": true,
    "mean": -0.4334,
    "std": 0.7597,
    "p05": -1.0,
    "p95": 1.0,
    "threshold": 0.02,
    "threshold_method": "otsu_rejected_prior_used",
    "separability": 0.872,
    "note": "Otsu suggested -0.11, outside the plausible EXG window [-0.08, +0.12] \u2014 the dominant histogram split is not this class boundary, so t
  },
  {
    "name": "TEXTURE",
    "available": true,
    "mean": 0.005,
    "std": 0.0038,
    "p05": 0.0005,
    "p95": 0.0121,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "local standard deviation of brightness, 5x5 window"
  },
  {
    "name": "MNDWI",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "requires Green and SWIR1 bands (absent from 4-band and RGB products)"
  },
  {
    "name": "NDBI",
    "available": false,
    "mean": null,
    "std": nu
11,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "requires SWIR1 and NIR bands; built-up detected from brightness and texture instead"
  },
  {
    "name": "NDVI",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "requires Red and NIR bands; product has blue, green, red"
  },
  {
    "name": "NDWI",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "requires Green and NIR bands"
  }
],
"regions": [
  {
    "label": "bare soil or fallow ground",
    "class_name": "bare_soil",
    "bbox_norm": [
      23.83,
      11.13,
      99.8,
      99.8
    ]
  }
],

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"bbox_px": [
  122,
  57,
  511,
  511
],
"centroid_norm": [
  70.32,
  66.55
],
"area_px": 55151,
"area_ha": 21.459,
"fraction": 0.21759,
"position": "eastern",
"compactness": 0.312,
"score": 0.725
},
{
  "label": "vegetation",
  "class_name": "vegetation",
  "bbox_norm": [
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    50.98,
    53.71,
    99.22
  ],
  "bbox_px": [
    118,
    261,
    275,
    508
  ],
  "centroid_norm": [
    38.68,
    74.95
  ],
  "area_px": 25169,
  "area_ha": 9.793,
  "fraction": 0.0993,
  "position": "south",
  "compactness": 0.649,
  "score": 0.829
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{
  "label": "water",
  "class_name": "water",
  "bbox_norm": [
    0.2,
    6.05,
    34.38,
    58.59
  ],
  "bbox_px": [
    1,
    31,
    176,
    300
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  "centroid_norm": [
    16.45,
    34.94
  ],
  "area_px": 22775,
  "area_ha": 8.862,
  "fraction": 0.08986,
  "position": "west",
  "compactness": 0.484,
  "score": 0.751
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  "class_name": "vegetation",
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    0.2,
    64.45,
    16.02
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    330,
    82
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  "centroid_norm": [
    26.22,
    5.23
  ],
  "area_px": 14983,
  "area_ha": 5.83,
  "fraction": 0.05911,
  "position": "north-west",
  "compactness": 0.562,
  "score": 0.746
},
{
  "label": "vegetation",
  "class_name": "vegetation",
  "bbox_norm": [
    69.73,
    20.31,
    99.8,

```

```

45.7
],
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357,
104,
511,
234
],
"centroid_norm": [
86.87,
33.43
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"area_ha": 4.995,
"fraction": 0.05065,
"position": "east",
"compactness": 0.641,
"score": 0.767
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{
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"class_name": "bare_soil",
"bbox_norm": [
41.8,
0.2,
80.08,
33.79
],
"bbox_px": [
214,
1,
410,
173
],
"centroid_norm": [
62.2,
13.52
],
"area_px": 12651,
"area_ha": 4.922,
"fraction": 0.04991,
"position": "north",
"compactness": 0.375,
"score": 0.66
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"class_name": "bare_soil",
"bbox_norm": [
0.2,
57.23,
26.95,
82.81
],
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1,
293,
138,
424
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11.74,
68.45
],
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"area_ha": 4.815,
"fraction": 0.04882,
"position": "south-west",
"compactness": 0.689,
"score": 0.784
},
{
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83.59,
24.8,
99.8
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1,
428,
127,
511
],
"centroid_norm": [
10.53,
93.29
],
"area_px": 7673,
"area_ha": 2.986,
"fraction": 0.03027,
"position": "south-west",
"compactness": 0.734,
"score": 0.78
},
{
"label": "built-up / man-made surface",
"class_name": "built_up",
"bbox_norm": [

```

```

67.38,
0.2,
90.82,
27.34
],
"bbox_px": [
345,
1,
465,
140
],
"centroid_norm": [
80.69,
16.02
],
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"area_ha": 2.063,
"fraction": 0.02092,
"position": "north-east",
"compactness": 0.318,
"score": 0.602
},
{
"label": "bare soil or fallow ground",
"class_name": "bare_soil",
"bbox_norm": [
56.05,
22.27,
79.88,
36.52
],
"bbox_px": [
287,
114,
409,
187
],
"centroid_norm": [
66.14,
28.15
],
"area_px": 5121,
"area_ha": 1.993,
"fraction": 0.0202,
"position": "north",
"compactness": 0.575,
"score": 0.704
},
{
"label": "water",
"class_name": "water",
"bbox_norm": [
87.3,
0.2,
99.8,
18.16
],
"bbox_px": [
447,
1,
511,
93
],
"centroid_norm": [
94.0,
8.65
],
"area_px": 5120,
"area_ha": 1.992,
"fraction": 0.0202,
"position": "north-east",
"compactness": 0.87,
"score": 0.822
},
{
"label": "built-up / man-made surface",
"class_name": "built_up",
"bbox_norm": [
51.37,
29.3,
78.71,
52.73
],
"bbox_px": [
263,
150,
403,
270
],
"centroid_norm": [
65.87,
40.61
],
"area_px": 4834,
"area_ha": 1.881,
"fraction": 0.01907,
"position": "centre",
"compactness": 0.288,
"score": 0.588
}
},
"quality": {

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"band_layout": "rgb",
"bands_available": [
  "blue",
  "green",
  "red"
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"indices_available": [
  "BRIGHTNESS",
  "BWI",
  "EXG",
  "TEXTURE"
],
"indices_unavailable": [
  "MNDWI",
  "NDBI",
  "NDVI",
  "NDWI"
],
"valid_pixel_fraction": 0.9669,
"separability": 0.8912,
"spatial_reference": true,
"limitations": [
  "No near-infrared band: NDVI/NDWI unavailable, vegetation and water detected from visible-band proxies (ExG, blue-over-red) with reduced reliability"
],
},
"scene_labels": [],
"label_source": null
},
"sar": {
  "kind": "land_cover",
  "method": "sar_backscatter_roughness"
},
s_v1",
"classes": [
  {
    "name": "water",
    "label": "water",
    "pixel_count": 3596,
    "fraction": 0.01383795432243665,
    "area_ha": 1.399200439453125,
    "evidence_basis": "backscatter < 16.9 dB (dark_mode_otsu, scene-relative (uncalibrated DN)) with low roughness (texture <= p45 = 0.73 dB): specular",
    "reliability": "medium",
    "region_count": 15
  },
  {
    "name": "vegetation",
    "label": "vegetation",
    "pixel_count": 0,
    "fraction": 0.0,
    "area_ha": 0.0,
    "evidence_basis": "not claimed: single-polarisation SAR carries no volume-scattering discriminator for vegetation",
    "reliability": "unavailable",
    "region_count": 0
  },
  {
    "name": "built_up",
    "label": "built-up / man-made surface",
    "pixel_count": 3,
    "fraction": 1.1544455775114002e-05,
    "area_ha": 0.00116729736328125,
    "evidence_basis": "backscatter > p75 (40.5 dB) with roughness > 1.60 dB: double-bounce returns adjacent to radar shadow",
    "reliability": "medium",
    "region_count": 0
  },
  {
    "name": "bare_soil",
    "label": "bare soil or fallow ground",
    "pixel_count": 110715,
    "fraction": 0.42604814038058225,
    "area_ha": 43.07910919189453,
    "evidence_basis": "low-roughness surface residue, consistent with bare or fallow ground",
    "reliability": "medium",
    "region_count": 30
  },
  {
    "name": "other",
    "label": "unclassified",
    "pixel_count": 147789,
    "fraction": 0.5687145248494411,
    "area_ha": 57.50457000732422,
    "evidence_basis": "intermediate backscatter and roughness, not assigned",
    "reliability": "medium",
    "region_count": 34
  }
],
"dominant": "bare_soil",
"indices": [
  {
    "name": "BRIGHTNESS",
    "available": true,
    "mean": 0.6602,
    "std": 0.2652,
    "p05": 0.2658,
    "p95": 0.9897,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "mean scaled reflectance across available bands"
  },
  {
    "name": "SAR_DB",
    "available": true,

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    "mean": 27.3994,
    "std": 14.4311,
    "p05": 6.423,
    "p95": 45.0838,
    "threshold": 16.8651,
    "threshold_method": "dark_mode_otsu",
    "separability": 0.6174,
    "note": "backscatter: darkest histogram mode split at 16.87 (scene-relative; the product carries an unknown gain)"
  },
  {
    "name": "SAR_TEXTURE",
    "available": true,
    "mean": 3.6561,
    "std": 5.8645,
    "p05": 0.3084,
    "p95": 16.8738,
    "threshold": 1.6,
    "threshold_method": "otsu_rejected_prior_used",
    "separability": 0.8971,
    "note": "Otsu suggested +8.84, outside the plausible SAR_TEXTURE window [+0.40, +2.80] \u2014 the dominant histogram split is not this class boundary"
  },
  {
    "name": "TEXTURE",
    "available": true,
    "mean": 0.0417,
    "std": 0.051,
    "p05": 0.0053,
    "p95": 0.1489,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "local standard deviation of brightness, 5x5 window"
  },
  {
    "name": "MNDWI",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "not defined for SAR imagery (no spectral bands)"
  },
  {
    "name": "NDBI",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "not defined for SAR imagery (no spectral bands)"
  },
  {
    "name": "NDVI",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "not defined for SAR imagery (no spectral bands)"
  },
  {
    "name": "NDWI",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "not defined for SAR imagery (no spectral bands)"
  },
  {
    "name": "SAR_RATIO",
    "available": false,
    "mean": null,
    "std": null,
    "p05": null,
    "p95": null,
    "threshold": null,
    "threshold_method": null,
    "separability": null,
    "note": "requires dual-polarisation data (VV+VH or HH+HV); single-pol product carries no polarimetric ratio"
  }
],
"regions": [
  {
    "label": "bare soil or fallow ground",
    "class_name": "bare_soil",
    "bbox_norm": [
      0.2,

```

```

0.2,
99.8,
65.04
],
"bbox_px": [
1,
1,
511,
333
],
"centroid_norm": [
43.96,
28.45
],
"area_px": 87130,
"area_ha": 33.902,
"fraction": 0.33529,
"position": "northern",
"compactness": 0.515,
"score": 0.806
},
{
"label": "bare soil or fallow ground",
"class_name": "bare_soil",
"bbox_norm": [
0.2,
65.82,
46.68,
99.8
],
"bbox_px": [
1,
337,
239,
511
],
"centroid_norm": [
25.96,
86.79
],
"area_px": 17747,
"area_ha": 6.905,
"fraction": 0.06829,
"position": "south-west",
"compactness": 0.429,
"score": 0.703
},
{
"label": "bare soil or fallow ground",
"class_name": "bare_soil",
"bbox_norm": [
21.68,
46.68,
27.73,
50.98
],
"bbox_px": [
111,
239,
142,
261
],
"centroid_norm": [
24.73,
48.89
],
"area_px": 277,
"area_ha": 0.108,
"fraction": 0.00107,
"position": "west",
"compactness": 0.406,
"score": 0.614
},
{
"label": "water",
"class_name": "water",
"bbox_norm": [
22.27,
53.52,
28.12,
56.64
],
"bbox_px": [
114,
274,
144,
290
],
"centroid_norm": [
24.96,
54.93
],
"area_px": 276,
"area_ha": 0.107,
"fraction": 0.00106,
"position": "west",
"compactness": 0.575,
"score": 0.681
},
{
"label": "water",
"class_name": "water",

```

```

    "bbox_norm": [
      75.0,
      48.63,
      83.79,
      51.17
    ],
    "bbox_px": [
      384,
      249,
      429,
      262
    ],
    "centroid_norm": [
      79.29,
      49.82
    ],
    "area_px": 276,
    "area_ha": 0.107,
    "fraction": 0.00106,
    "position": "east",
    "compa
ctness": 0.472,
    "score": 0.64
  },
  {
    "label": "bare soil or fallow ground",
    "class_name": "bare_soil",
    "bbox_norm": [
      76.37,
      22.07,
      83.01,
      25.98
    ],
    "bbox_px": [
      391,
      113,
      425,
      133
    ],
    "centroid_norm": [
      79.78,
      23.81
    ],
    "area_px": 254,
    "area_ha": 0.099,
    "fraction": 0.00098,
    "position": "north-east",
    "compactness": 0.374,
    "score": 0.601
  },
  {
    "label": "water",
    "class_name": "water",
    "bbox_norm": [
      89.06,
      13.87,
      93.95,
      16.99
    ],
    "bbox_px": [
      456,
      71,
      481,
      87
    ],
    "centroid_norm": [
      91.33,
      15.22
    ],
    "area_px": 224,
    "area_ha": 0.087,
    "fraction": 0.00086,
    "position": "north-east",
    "compactness": 0.56,
    "score": 0.675
  },
  {
    "label": "water",
    "class_name": "water",
    "bbox_norm": [
      87.7,
      0.2,
      90.23,
      4.3
    ],
    "bbox_px": [
      449,
      1,
      462,
      22
    ],
    "centroid_norm": [
      88.73,
      1.87
    ],
    "area_px": 214,
    "area_ha": 0.083,
    "fraction": 0.00082,
    "position": "north-east",
    "compactness": 0.784,
    "score": 0.765
  }

```

```

    }
  ],
  "quality": {
    "band_layout": "sar_single",
    "bands_available": [
      "amplitude"
    ],
    "indices_available": [
      "BRIGHTNESS",
      "SAR_DB",
      "SAR_TEXTURE",
      "TEXTURE"
    ],
    "indices_unavailable": [
      "MNDWI",
      "NDBI",
      "NDVI",
      "NDWI",
      "SAR_RATIO"
    ],
    "valid_pixel_fraction": 0.9913,
    "separability": 0.7573,
    "spatial_reference": true,
    "limitations": [
      "SAR-only input: classes derive from backscatter level and roughness; vegetation vigour and water chemistry are not observable."
    ]
  },
  "scene_labels": [],
  "label_source": null
},
"overlay_path": "C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_fusion.png",
"quality": {
  "band_layout": "optical:rgb + sar:sar_single",
  "bands_available": [
    "blue",
    "green",
    "red",
    "sar:amplitude"
  ],
  "indices_available": [
    "BRIGHTNESS",
    "BWI",
    "EXG",
    "SAR_DB",
    "SAR_TEXTURE",
    "TEXTURE"
  ],
  "indices_unavailable": [
    "MNDWI",
    "NDBI",
    "NDVI",
    "NDWI"
  ],
  "valid_pixel_fraction": 0.959,
  "separability": 0.8912,
  "spatial_reference": true,
  "limitations": [
    "No near-infrared band: NDVI/NDWI unavailable, vegetation and water detected from visible-band proxies (ExG, blue-over-red) with reduced reliability",
    "SAR-only input: classes derive from backscatter level and roughness; vegetation vigour and water chemistry are not observable.",
    "0.0% of the optical scene is bright, smooth and non-vegetated (brightness >= 0.85): treated as cloud/haze obscured"
  ]
}
},
"outputs": {
  "text": {
    "text": "Based on the measured signal, using the optical and SAR images jointly: built-up / man-made surface 10.7% (10.5 ha); water-covered 12.3% (12.3 ha).",
    "narration_source": "mock",
    "n_samples": 1
  },
  "boxes": [
    {
      "label": "water",
      "bbox": [
        0.2,
        6.05,
        34.38,
        58.59
      ],
      "score": 0.751,
      "class_name": "water",
      "area_ha": 8.822,
      "position": "west"
    },
    {
      "label": "built-up / man-made surface",
      "bbox": [
        67.38,
        0.2,
        90.82,
        27.34
      ],
      "score": 0.602,
      "class_name": "built-up",
      "area_ha": 2.052,
      "position": "north-east"
    },
    {
      "label": "water",
      "bbox": [

```

```

87.3,
0.2,
99.8,
18.16
],
"score": 0.815,
"class_name": "water",
"area_ha": 1.951,
"position": "north-east"
},
{
"label": "built-up / man-made surface",
"bbox": [
51.37,
29.3,
78.71,
52.73
],
"score": 0.588,
"class_name": "built_up",
"area_ha": 1.876,
"position": "centre"
},
{
"label": "built-up / man-made surface",
"bbox": [
6.45,
79.69,
32.42,
99.8
],
"score": 0.609,
"class_name": "built_up",
"area_ha": 1.823,
"position": "south-west"
},
{
"label": "water",
"bbox": [
43.36,
59.77,
67.97,
74.22
],
"score": 0.608,
"class_name": "water",
"area_ha": 1.291,
"position": "south"
},
{
"label": "built-up / man-made surface",
"bbox": [
36.91,
8.01,
49.61,
39.26
],
"score": 0.59,
"class_name": "built_up",
"area_ha": 1.261,
"position": "north"
}
],
"mask": {
"path": "C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_fusion.png",
"kind": "fusion_map",
"changed_fraction": null
},
"evidence": [
{
"role": "input_preview",
"path": "C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_areaB_optical.png",
"caption": "optical \u00b7 areaB_optical.tif"
},
{
"role": "input_preview",
"path": "C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_areaB_sar.png",
"caption": "SAR \u00b7 areaB_sar.tif (amplitude)"
},
{
"role": "side_by_side",
"path": "C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_fusion_pair.png",
"caption": "Co-registered optical / SAR pair after SAR calibration"
},
{
"role": "fusion_map",
"path": "C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_fusion.png",
"caption": "Joint optical+SAR classes; white outline marks optically obscured area"
}
],
"evidence_thumbnails": [
"C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_areaB_optical.png",
"C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_areaB_sar.png",
"C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_fusion_pair.png",
"C:\\project\\satquery\\backend\\runtime\\outputs\\job_1a90dc430a12_fusion.png"
]
},
"confidence": {
"value": null,
"source": "mock",
"calibrated": false,
"components": [],

```

```
    "note": "Mock backend was active (SATQUERY MOCK=1): no measurement was performed, so no confidence is reported."
  },
  "execution_time_ms": 1374,
  "model_backend": "mock",
  "environment": {
    "server_version": "0.2.0",
    "device": "cpu",
    "torch_available": true,
    "vlm_narration_enabled": false,
    "vlm_model_id": null,
    "mock_mode": true,
    "max_analysis_pixels": 4194304
  },
  "error": null,
  "created_at": "20
26-08-30T16:58:39.700176+00:00",
  "server_version": "0.2.0"
}
```